

Multi-Gas Greenhouse Emissions Forecasting and Climate Risk Modeling System

1. Introduction

Greenhouse gas emissions are one of the most important drivers of climate change. The three major gases—carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O)—contribute differently to global warming due to their varying atmospheric lifetimes and radiative forcing strengths. CO₂ is the dominant long-term driver, CH₄ has a strong short-term warming effect, and N₂O is highly persistent with a significant warming impact per molecule.

This project focuses on building a predictive system that models and forecasts these three gases using historical emissions data. The goal is not only to forecast emissions but also to understand climate dynamics and build a simplified climate risk assessment system.

2. Problem Statement

Most emissions analysis focuses only on CO₂, which provides an incomplete view of climate change. This is a limitation because:

- Methane (CH₄) has a much higher warming potential in the short term.
- Nitrous oxide (N₂O) has long atmospheric persistence and strong radiative forcing.
- Different gases originate from different economic sectors such as agriculture, energy, and industry.

Therefore, the problem addressed is how to jointly model and forecast multiple greenhouse gases to better understand future climate risk and system behavior.

3. Objectives

The main objective is to develop a multi-gas forecasting system for climate analysis.

Specific objectives include:

- Clean and structure multi-country emissions data.
- Build time-series forecasting models for CO₂, CH₄, and N₂O.
- Analyze trends, seasonality, and stationarity in emissions data.
- Convert emissions into a unified CO₂-equivalent climate metric.
- Generate future projections for emissions under historical patterns.

4. Dataset Description

The dataset used contains historical emissions data across countries and sectors from 1970 to 2018. It includes:

- Country names and ISO codes
- Sector classifications (energy, transport, industry, agriculture, etc.)
- Fossil and bio-based emissions separation
- Yearly emissions values represented as columns (Y_1970 to Y_2018)

The dataset is transformed from a wide format into a long format to enable time-series modeling.

5. Methodology

5.1 Data Preprocessing

The dataset is cleaned by:

- Removing missing or duplicate values
 - Standardizing column formats
 - Converting year-based columns into numeric time indices
 - Aggregating emissions at country and sector levels
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5.2 Time-Series Construction

Separate time series are constructed for each gas:

- CO₂ emissions per year
- CH₄ emissions per year
- N₂O emissions per year

Each series represents total emissions per country per year.

5.3 Stationarity Testing

The Augmented Dickey-Fuller (ADF) test is used to check stationarity. Results show that emissions data is non-stationary due to strong upward trends over time. First-order differencing is applied to stabilize the series.

5.4 Temporal Dependency Analysis

Autocorrelation analysis shows:

- Strong correlation at short lags (1–3 years)
- Gradual decay after lag 3
- Weak correlation beyond lag 5

This indicates that emissions are highly dependent on recent historical values rather than distant past values.

5.5 Forecasting Model

The primary model used is SARIMA (Seasonal Autoregressive Integrated Moving Average). It captures:

- Trend components using differencing
- Autoregressive structure from past values
- Moving average components from past errors
- Weak seasonal patterns in emissions data

Each gas is modeled independently using SARIMA.

6. Climate Impact Modeling

To unify multiple gases into a single climate measure, emissions are converted into CO₂-equivalent (CO₂e). This uses global warming potential (GWP) weighting:

- CH₄ has a significantly higher warming effect than CO₂ in the short term
- N₂O has the highest long-term impact per unit

This allows all gases to be compared on a common scale and supports climate risk analysis.

7. System Outputs

The system produces:

- Forecasted emissions for CO₂, CH₄, and N₂O
- Sector-level emission trends
- CO₂-equivalent total emissions
- Growth rate analysis over time

- Climate risk indicators based on emission trajectories
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8. Key Findings

- CO₂ emissions show strong long-term upward trends driven by industrial activity.
 - CH₄ and N₂O introduce additional variability and complexity in climate behavior.
 - Emissions are highly dependent on recent historical values.
 - The system is more trend-driven than shock-driven.
 - Multi-gas modeling provides a more complete understanding of climate dynamics compared to single-gas analysis.
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9. Limitations

- SARIMA assumes linear relationships and may not capture complex nonlinear climate behavior.
 - External factors such as policy changes, wars, or economic shocks are not explicitly modeled.
 - Forecast accuracy depends heavily on historical data quality.
 - Structural breaks in emissions trends may reduce model accuracy.
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10. Conclusion

This project demonstrates that greenhouse gas emissions can be modeled as structured time-dependent systems. By extending analysis beyond CO₂ to include CH₄ and N₂O, the model provides a more complete representation of climate forcing dynamics.

The system moves beyond simple forecasting and becomes a climate risk analysis framework capable of supporting environmental research and policy evaluation.

11. Future Work

Future improvements include:

- Multi-variable models such as VAR for joint gas forecasting
- Deep learning models such as LSTM for nonlinear pattern learning
- Integration of economic and energy indicators as external variables
- Scenario-based forecasting (policy simulations)
- Deployment as an interactive dashboard for real-time climate analysis